

Evaluation Metrics

	Accuracy In Interview	Accuracy
SVM-classifier	What is the overall performance of the Support Vector Machine model	0.79
DT-Classifier	What is the overall performance of the Decision Tree model	0.91
RF-Classifier	What is the overall performance of the Random Forest model	0.91

Recall

	What is the correct classification of not purchased (0)/ What percentage of actual purchasers were correctly classified as not purchased?	What is the correct classification of purchased(1)/ What percentage of actual purchasers were correctly classified as not purchased(1)?
Recall		
SVM-classifier	0.97	0.44
DT-Classifier	0.92	0.88
RF-Classifier	0.9	0.93

Precision

	What percentage of customers predicted as purchased actually not purchased (0) ?	What percentage of customers predicted as purchased actually purchased (1) ?
Precision		
SVM-classifier	0.77	0.9
DT-Classifier	0.96	0.83
RF-Classifier	0.94	0.86

F1-Score

	1.How effectively does the model identify the positive class while minimizing false positives and false negatives? 2. How well are Precision and Recall balanced? 3.Suppose your model has high Precision but low Recall. Which metric would you use to evaluate the overall performance? 4.What is the combined measure of Precision and Recall? => (Not Purchased)	1.How effectively does the model identify the positive class while minimizing false positives and false negatives? 2. How well are Precision and Recall balanced? 3.Suppose your model has high Precision but low Recall. Which metric would you use to evaluate the overall performance? 4.What is the combined measure of Precision and Recall? => (Purchased)
F1-Score		

SVM-classifier	0.86	0.59
DT-Classifier	0.93	0.87
RF-Classifier	0.93	0.87
Support		
Support	What is Support in Machine Learning? Support is the number of actual occurrences (true instances) of each class in the dataset. It is simply the count of samples belonging to each class. =>Not Purchased	What is Support in Machine Learning? (Purchased)
SVM-classifier	79	41
DT-Classifier	79	41
RF-Classifier	79	41

	1.How does the model perform on average across all classes without considering class imbalance? 2.What is the average Precision across all classes, giving equal importance to each class? 3.What is the average Recall across all classes, treating every class equally? 4.What is the average F1-score across all classes, where each class contributes equally?				
Macro Average			Precision	Recall	F1-Score
		SVM-classifier	0.83	0.71	0.73
		DT-Classifier	0.89	0.91	0.9
		RF-Classifier	0.9	0.9	0.9

Weighted Average	1.What is the average Precision across all classes, weighted by the number of samples in each class?				
	2.What is the average Recall across all classes, considering the number of instances in each class?				
	3.What is the average F1-score across all classes, weighted by class support?				
	4.How does the model perform across all classes when larger classes are given more importance than smaller classes?		Precision	Recall	F1-Score
		SVM-classifier	0.81	0.79	0.77
		DT-Classifier	0.91	0.91	0.91
		RF-Classifier	0.91	0.91	0.91